

Thematic Categorization of Fake News in Online News Outlets

Abstract

The proliferation of fake news poses significant challenges to societal stability, influencing public opinion and undermining democratic processes. This study investigates the prevalence of fake news across various domains in online news platforms. Utilizing datasets from six news websites and established repositories, we employed machine learning models—Naïve Bayes, Support Vector Machine (SVM), and Long Short-Term Memory (LSTM) neural networks—for binary and multi-class classification tasks. The LSTM model outperformed others, achieving 99.72% accuracy in fake news detection and 84.6% in topic classification. Findings indicate that fake news is most prevalent in the political domain. The study also integrates recent research (2020–2025), offering a robust and current theoretical foundation. It highlights the importance of down-weighting high-frequency keywords during significant events, thereby enhancing classification performance.

Keywords: fake news detection, machine learning, LSTM, deep learning, text classification, online news

1. Introduction

The rapid proliferation of fake news in the digital age poses a significant threat to public trust, social cohesion, and democratic governance. With the ubiquity of internet access and the exponential growth of social media platforms, misinformation can now spread faster and more widely than ever before. Unlike traditional media, which is typically subject to editorial oversight, online news ecosystems often lack quality control mechanisms, making them highly susceptible to the dissemination of fabricated or misleading content.

Fake news, broadly defined as deliberately false or misleading information presented as news, has become a global issue. It has been weaponized to influence elections, undermine public health measures, distort economic markets, and incite social unrest. The 2016 U.S. presidential election and the COVID-19 pandemic are two prominent examples where fake news had tangible and far-reaching impacts. These events underscored the urgency of developing reliable and scalable methods for identifying and mitigating misinformation.

Numerous approaches have been proposed to detect fake news, ranging from manual fact-checking by professional organizations to automated classification using artificial intelligence. While fact-checking is thorough, it is slow and resource-intensive. In contrast, machine learning and natural language processing (NLP) offer promising alternatives that can analyze large volumes of data quickly and effectively. Yet, many existing models struggle with generalization across different topics, languages, and writing styles.

This study aims to bridge these gaps by exploring both the detection and thematic categorization of fake news. Specifically, we investigate which domains—such as politics, health, business, and entertainment—are most vulnerable to fake news and how various machine

learning models perform in detecting it. We evaluate three classifiers—Naïve Bayes, Support Vector Machine (SVM), and Long Short-Term Memory (LSTM) neural networks—on datasets collected from reputable and unreliable news sources, including ISOT, HuffPost, and a custom-scraped 2020 dataset.

By analyzing linguistic patterns, topical focus, and classification outcomes, this research contributes to a deeper understanding of how fake news manifests across different thematic domains. The findings offer actionable insights for developing more adaptive and context-aware detection systems, which are crucial for enhancing media literacy and safeguarding public discourse in a digitally connected world.

2. Previous Studies

The issue of fake news has prompted significant scholarly interest over the past decade, spanning disciplines such as computer science, communication studies, psychology, and journalism. Researchers have proposed a variety of theoretical frameworks and empirical models to understand and address the spread of misinformation. In this section, we outline the evolution of fake news detection methodologies from traditional statistical techniques to advanced deep learning and transformer-based models, including recent developments from 2020 to 2025.

2.1 Traditional Machine Learning Approaches

Early efforts to detect fake news relied on feature-based models using manually engineered attributes derived from the text. Reddy (2019) applied text-mining techniques that extracted term frequency-inverse document frequency (TF-IDF) vectors to feed into a Naïve Bayes classifier, achieving 94.49% accuracy. Henrique (2020) introduced a platform-independent classifier that focused on lexical and stylistic features, capable of functioning across languages and platforms.

Bhardwaj et al. (2019) investigated a variety of algorithms including Random Forests, Decision Trees, and Naïve Bayes. Their study found that while Naïve Bayes was efficient, ensemble models like Random Forests provided higher stability. These models often relied on bag-of-words or n-gram features, which limited their ability to capture context or long-term dependencies in text.

2.2 Neural Networks and Deep Learning

With the rise of neural networks, researchers began exploring architectures capable of understanding text at a deeper, semantic level. Zheng (2020) introduced "Fake Detector," a deep propagation model combining content and social context. Gravvanis et al. (2019) constructed the UNB dataset and applied LSTM networks, demonstrating improved classification accuracy over traditional models.

Deep learning models like Convolutional Neural Networks (CNNs), Recurrent Neural Networks (RNNs), and their variants such as LSTMs and GRUs have been widely adopted for their capacity to model sequential and hierarchical structures in text. These models have consistently

outperformed traditional approaches, particularly in handling nuanced language, sarcasm, and informal expressions often present in fake news.

2.3 Social and Psychological Dimensions

Beyond technical detection methods, scholars have also examined the psychological and sociological mechanisms behind the spread of fake news. Jang and Kim (2018) investigated third-person perception and found that individuals who believe others are more susceptible to fake news are more likely to support media literacy interventions. Waszak et al. (2018) examined health misinformation in Polish social media, concluding that public susceptibility to fake news is strongly influenced by trust in institutions and perceived credibility of sources.

2.4 Post-2020 Advances and Transformer Models

The years following 2020 witnessed a surge in research focusing on transformer-based models, interpretability, and fairness in fake news detection. Zhou et al. (2021) proposed a BERT-based architecture with attention mechanisms that improved classification performance by learning contextual embeddings of words. Kaliyar et al. (2021) introduced FakeBERT, a multi-modal model integrating textual and visual data, which outperformed single-modal models on multiple benchmark datasets.

Patwa et al. (2021) provided a comprehensive review of fake news research during the COVID-19 pandemic, highlighting the unique challenges of misinformation during public health crises. Wang et al. (2022) presented a temporal-aware fake news detection model capable of tracking the evolution of linguistic patterns and adapting to changing narratives.

Sharma and Shadab (2023) developed an interpretable hybrid attention-based model that allowed for both high accuracy and transparency, essential for deployment in real-world applications. Alam et al. (2024) tackled the issue of bias in fake news datasets, identifying demographic and topic-based disparities and suggesting methods to mitigate their impact on model fairness.

Collectively, these studies illustrate the multifaceted nature of fake news research. From statistical methods and deep learning to interdisciplinary approaches that incorporate social context and psychological theory, the field continues to evolve rapidly. Our study builds upon this foundation by combining classification performance with topic-level categorization, offering a more nuanced understanding of fake news distribution across different domains.

3. Data and Methodology

3.1 Data Sources

This study uses three major datasets that collectively span multiple regions, topics, and journalistic standards, providing a robust and diverse foundation for fake news analysis:

- **ISOT Fake News Dataset:** Provided by the University of Victoria, this dataset contains 21,417 real and 23,481 fake news articles. Real articles were sourced from Reuters,

while fake ones were collected from unreliable sources like World News Politics. Each entry includes a headline and body text, making it ideal for binary classification tasks.

- **2020 Scraped News Data:** Using the Octoparse web-scraping tool, we collected articles from six mainstream and partisan websites: CNN, USA Today, Newsmax, Dawn, The Scotsman, and Iran's Young Journalists Club (YJC). These sources were selected to represent a wide spectrum of media bias and credibility. The final scraped dataset includes thousands of English-language news stories spanning categories such as politics, health, economy, and world affairs.
- **HuffPost Dataset:** Originally containing 200,853 news articles labeled across 41 categories, we consolidated them into four high-level themes—Politics, Health, Business & Science, and Sports & Entertainment. Due to the imbalance across classes, we used SMOTE (Synthetic Minority Over-sampling Technique) to generate synthetic examples for underrepresented categories. The resulting balanced dataset significantly improved multi-class classifier performance.

3.2 Preprocessing

Before feeding the data into machine learning models, we implemented the following preprocessing steps:

- **Text Normalization:** Lowercasing all text to maintain uniformity.
- **HTML and Noise Removal:** Stripped HTML tags and metadata using BeautifulSoup.
- **Tokenization:** Divided sentences into individual words using NLTK.
- **Stopword Removal:** Filtered out high-frequency, low-importance words such as "the," "and," and "said."
- **Punctuation and Digit Cleaning:** Removed symbols and numbers unless contextually relevant.
- **Stemming and Lemmatization:** Standardized word forms to their base or root form for consistency across variations (e.g., "running" → "run").
- **Concatenation:** Headline and body were merged into a single text input to provide context-aware sequences for training.

3.3 Feature Engineering

While deep learning models like LSTM automatically extract features, traditional models (Naïve Bayes and SVM) require manual vectorization:

- **TF-IDF Vectorization:** Applied to Naïve Bayes and SVM to capture term frequency weighted by inverse document frequency.
- **Unigrams and Bigrams:** Used to detect common phrases and word pairs.

- **Keyword Weighting:** Terms like “coronavirus,” “election,” and “Trump” were tested for contextual relevance to avoid overfitting on topical spikes.

3.4 Model Architectures

- **Naïve Bayes:** A probabilistic classifier assuming feature independence. We used MultinomialNB due to its effectiveness with text data.
- **Support Vector Machine (SVM):** Implemented with both default and tuned parameters ($C=100$) and a linear kernel. This model was particularly effective in high-dimensional vector space but required significant computational time.
- **Long Short-Term Memory (LSTM):** The core model used for deep learning, designed to capture long-range dependencies in text. Our LSTM had over 168,000 trainable parameters and was trained using ReLU activation, dropout layers, and an Adam optimizer. It was trained for 10 epochs for binary classification and 22 for multi-class.

3.5 Evaluation Metrics

We used a comprehensive set of metrics to assess model performance:

- **Accuracy:** The overall correctness of predictions.
- **Precision:** The proportion of predicted positives that are truly positive.
- **Recall (Sensitivity):** The ability to find all actual positives.
- **F1-score:** The harmonic mean of precision and recall.
- **Confusion Matrix:** Visualized misclassification distribution across true vs. predicted labels.
- **Macro and Weighted Averages:** Especially relevant for imbalanced multi-class data.

These methodological strategies ensured fairness in comparison and robustness in evaluation, providing a reliable basis for drawing domain-specific conclusions.

4. Results

4.1 Binary Classification with ISOT Dataset

To evaluate fake news detection performance, we trained three classification models—Naïve Bayes, Support Vector Machine (SVM), and Long Short-Term Memory (LSTM)—on the ISOT dataset. This dataset includes a balanced number of real and fake news articles, making it suitable for binary classification tasks. The evaluation metrics included true positives (TP), true negatives (TN), false positives (FP), false negatives (FN), as well as accuracy and F1-score.

Table 1
Binary Classification Results on the ISOT Dataset

Model	TP	FP	FN	TN	Accuracy (%)	F1 Score (%)
Naïve Bayes	5122	278	245	5580	95.0	95.0
SVM (C=100)	5349	24	18	5834	99.4	99.5
LSTM	5354	18	13	5840	99.72	99.6

Note: TP = True Positives, FP = False Positives, FN = False Negatives, TN = True Negatives. LSTM model yielded the best results overall.

The Naïve Bayes model, though quick and interpretable, struggled to capture complex semantic relationships in the text. It showed high precision but relatively lower recall, misclassifying a significant number of nuanced or sarcastic fake articles due to its bag-of-words assumption.

The SVM model, equipped with a polynomial kernel and a tuned C-value, demonstrated a sharp increase in performance. It achieved 99.4% accuracy by learning from the high-dimensional feature space, though it required more computation time and fine-tuning.

The LSTM model emerged as the most effective, achieving 99.72% accuracy and a 99.6% F1-score. Its ability to recognize sequential dependencies and capture contextual flow in language made it far more resilient to subtle manipulations and linguistic ambiguities. For instance, it performed better on articles with misleading headlines but factual bodies or vice versa, where traditional models tended to misfire.

A detailed look at the confusion matrix shows only 13 fake articles were incorrectly classified as real, and 18 real articles were mislabeled as fake—showing extremely low false-positive and false-negative rates. This low misclassification rate affirms the LSTM’s robustness for deployment in sensitive, high-risk misinformation detection tasks.

4.2 Multi-class Classification with HuffPost Dataset

To explore thematic categorization, we applied the same models to the HuffPost dataset, reduced to four primary domains: Politics, Health, Business & Science, and Sports & Entertainment. Because the dataset was initially imbalanced—with politics having far more samples—we used the Synthetic Minority Over-sampling Technique (SMOTE) to ensure a fair distribution.

Table 2
Article Distribution per Topic Category (HuffPost Dataset, post-SMOTE)

Topic Category	Number of Articles
Politics	14,140
Business & Science	10,588
Sports & Entertainment	11,405
Health	10,845

Note: SMOTE was used to balance classes for improved performance in multi-class classification.

Once again, the LSTM model significantly outperformed both Naïve Bayes and SVM, achieving an accuracy of 84.6% and a macro-averaged F1-score of 77%. The improvement was particularly notable in underrepresented classes like Health and Business & Science, where traditional models failed to generalize well.

Error Analysis

- Naïve Bayes and SVM misclassified health articles as politics due to overlapping COVID-19-related vocabulary (“government mandates”, “policy change”, etc.).
- The LSTM model handled such overlap effectively by recognizing sequence-level patterns and sentence structure rather than relying solely on keyword frequency.

Key Observations by Topic

- **Politics:** This category contained the largest volume of fake news, driven by the U.S. presidential election, partisan controversies, and policy debates. Common patterns included accusations of fraud, manipulated poll data, and sensational claims about political figures. Language in this category often included emotionally charged words and references to divisive themes like nationalism or immigration.
- **Health:** Dominated by misinformation around COVID-19, including anti-vaccine rhetoric, unproven treatments, and conspiracy theories. This content often used pseudo-scientific terms to mimic credibility. LSTM models were particularly good at distinguishing legitimate scientific articles from fabricated ones due to their sensitivity to context flow.
- **Business & Science:** Articles in this category exaggerated economic trends or misrepresented tech innovations. Fake articles sometimes mirrored press releases or research briefs but introduced distortion through cherry-picked data or unfounded extrapolations. Misleading stock forecasts and fake startup announcements were common.
- **Sports & Entertainment:** This domain had the least volume but showed high virality. Fake stories often included celebrity deaths, relationship rumors, or event cancellations. While less harmful in content, such articles generated high engagement and reflected how misinformation can thrive even in casual consumption settings.

Strategic Insight

The results confirm that fake news varies dramatically by topic—not only in content but also in linguistic structure, emotional tone, and virality triggers. Political and health fake news are especially problematic due to their societal implications and the complexity of language used.

This affirms the need for topic-aware fake news detection systems—models that adapt their feature extraction and classification logic based on the article’s domain. For example:

- Health-related fake news detectors may need integration with verified medical databases.
- Political misinformation detectors might benefit from stance detection and sentiment analysis.

These findings highlight a critical opportunity: future fake news models should not be general-purpose but rather modular and context-sensitive, capable of adjusting their detection strategies based on content domain.

5. Discussion

The results of this study highlight the complex and domain-specific nature of fake news content. Political fake news dominated both in volume and influence, as evidenced by its prevalence across multiple datasets and its frequent targeting of prominent figures and divisive issues. This suggests that political content is uniquely vulnerable to misinformation campaigns, often due to its emotional charge, polarizing topics, and the high stakes involved in public opinion formation.

Health-related fake news surged in 2020 primarily due to the COVID-19 pandemic. The rapid evolution of scientific knowledge, mixed messaging from authorities, and heightened public fear created an environment ripe for the spread of false or misleading health information. The frequent use of scientific-sounding language in fake health news suggests a need for classifiers that can distinguish between legitimate technical discourse and pseudo-scientific jargon.

The performance differences among the models underscore the importance of model selection and data preparation in fake news detection. While Naïve Bayes offered acceptable performance, it lacked the ability to capture deeper semantic relationships. SVM improved on this by leveraging hyperplane optimization in a high-dimensional space. However, LSTM’s ability to learn long-term dependencies and sequential context gave it a decisive edge in both binary and multi-class classification tasks.

The use of SMOTE for class balancing was instrumental in improving multi-class classification performance. Without it, the underrepresented classes (particularly Health and Business & Science) suffered from low recall and poor generalization. This reinforces the importance of dataset engineering alongside model architecture.

These findings have practical implications for building real-world fake news detection systems. Future systems should incorporate topic-aware architectures, dynamic weighting for frequently appearing event-specific terms, and mechanisms for visual or multimedia content analysis.

Moreover, the use of explainable AI tools such as SHAP and LIME can further enhance user trust by offering transparency in prediction rationale, particularly in sensitive domains like politics and health.

6. Conclusion

This study has demonstrated the effectiveness of deep learning—specifically Long Short-Term Memory (LSTM) neural networks—in detecting and classifying fake news across different thematic domains. By leveraging comprehensive datasets and integrating recent advancements in machine learning and natural language processing, we were able to produce a robust model capable of achieving exceptional accuracy, especially in politically charged and health-related contexts.

The findings reaffirm that fake news is not randomly distributed but tends to cluster around socially and emotionally impactful topics. Political misinformation, for example, thrives on controversy and public engagement, while health misinformation leverages fear and scientific uncertainty. Recognizing these domain-specific patterns is key to improving future detection tools.

In comparing model performances, LSTM consistently outperformed traditional models like Naïve Bayes and SVM in both binary and multi-class tasks. This validates the utility of sequential learning architectures in processing complex language structures that characterize fake news.

Additionally, our use of SMOTE proved valuable in handling class imbalances that would otherwise bias models toward majority classes. Coupled with visualizations and n-gram analysis, our results contribute both quantitatively and qualitatively to the fake news detection literature.

Future work should aim to:

- Incorporate transformer-based models such as BERT and RoBERTa for enhanced contextual understanding.
- Extend detection capabilities to multimedia content including video and images.
- Improve model interpretability through explainable AI frameworks.
- Partner with fact-checking organizations to deploy real-time detection tools.

By combining technical rigor with practical insight, this research lays the groundwork for more sophisticated, transparent, and domain-sensitive fake news detection systems that are capable of adapting to the ever-changing media landscape.

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